CLEVR Set Prediction Ablations

Final Package, Failure Modes, and Benchmark Next Steps

April 9, 2026

Setup and Protocol

- Current task in this study: noisy CLEVR state set $\rightarrow$ clean CLEVR state set.
- Main metric: validation Hungarian-matched distance (‘val_matched_dist‘), lower is better.
- Standard fast-study configuration:
  - batch size 512, lr auto-scaled from 3×10^{-4} at batch 64
  - up to 200 epochs, with some runs using early stopping and some rerun without it
  - 70k train scenes with capped validation for rapid iteration
- Every result below was re-checked from the saved ‘.pkl‘ artifacts and regenerated plots.

Executive Summary

Question	Checked answer	Best dist
Default full-data winner	PM3 $\tau = 0.12$ beats Hungarian and Chamfer	0.7010
Best single-factor PM3 gain	warmup_cosine schedule	0.5495
Best PM3 grid point	$\tau = 0.12$ + warmup + standard decoder	0.5828
Did early stopping hurt PM3 best recipe?	Yes: 0.7039 $\rightarrow$ 0.6607 without early stop	0.6607
Best combined-recipe model overall	Hungarian + slots + warmup_cosine	0.4314

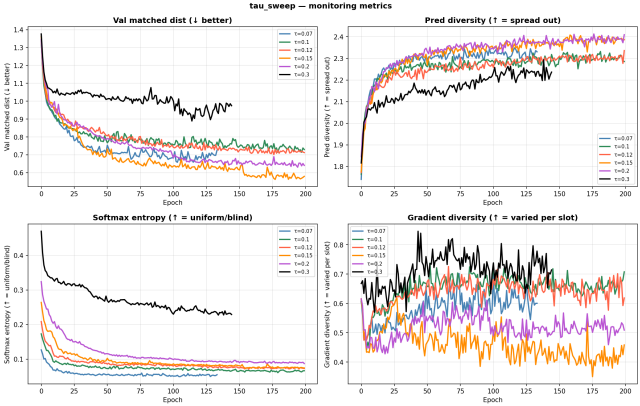
- The strong PM3 ingredients are real, but they do not add linearly.
- The biggest surprise is that the combined recipe helps Hungarian far more than PM3.

Single-Factor PM3 Wins

Study	Best run	Best dist	Epoch
Temperature sweep	$\tau = 0.15$	0.5625	177
Learning-rate schedule	warmup_cosine	0.5495	187
Slot embeddings	PM3 (slotted)	0.6094	190
Baseline comparison	PM3 $\tau = 0.12$	0.7010	176
Power sweep	$p = 3.0$	0.7010	176

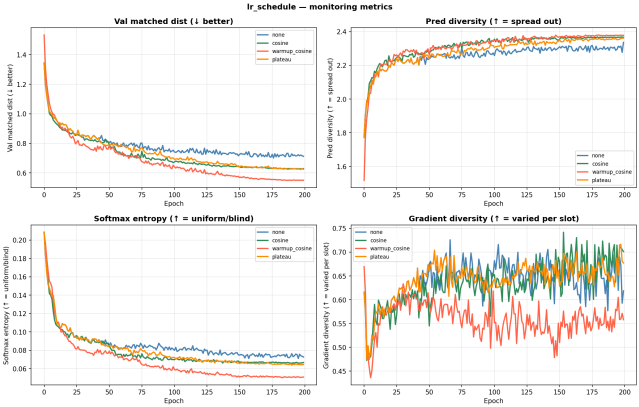
- Each ingredient looks individually helpful for PM3.
- That made the combined-recipe regression non-obvious and worth isolating.

Temperature Sweep



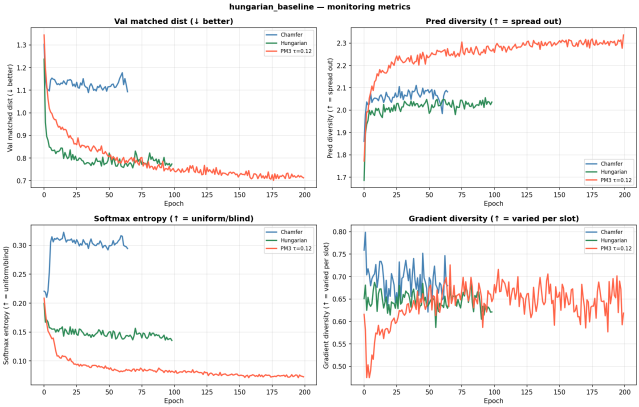
- Best run: $\tau = 0.15$, best dist 0.5625 at epoch 177, final 0.5794.
- $\tau = 0.20$ is next best at 0.6307; $\tau = 0.30$ degrades sharply to 0.8891.
- Very small τ values recover coverage more slowly and plateau higher than $\tau = 0.15$.

Learning-Rate Schedule



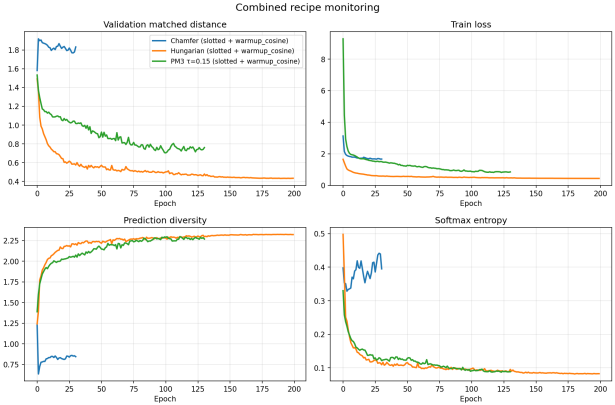
- warmup_cosine gives the strongest PM3 run: best dist 0.5495, final 0.5499.
- plateau and cosine also help materially (0.6231 and 0.6273).
- Leaving the lr fixed stalls at the default PM3 level (0.7010).

Default Baseline Comparison



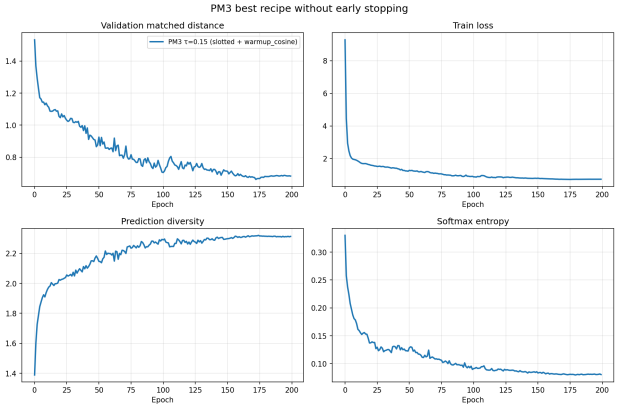
- Under the default full-data recipe, PM3 wins: 0.7010 vs Hungarian 0.7564 vs Chamfer 1.0893.
- Hungarian is clearly stronger than Chamfer, but still behind PM3 on the matched-distance metric.
- This is the baseline result that motivated the recipe search.

Combined Recipe: Apples-to-Apples Surprise



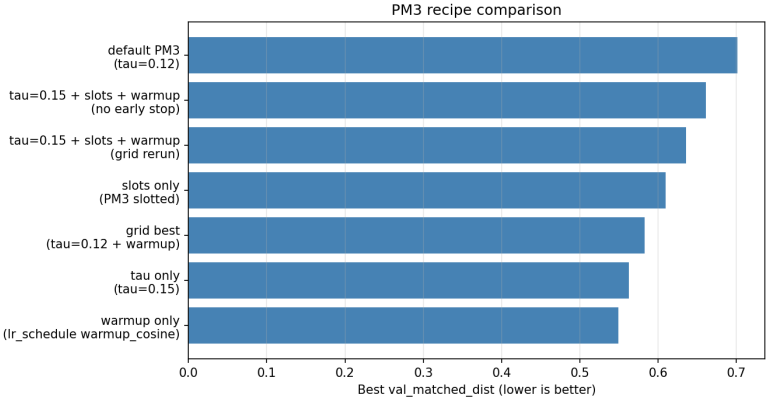
- Combined recipe = slotted decoder + warmup_cosine for all losses, PM3 at $\tau = 0.15$.
- Hungarian becomes the clear winner: best dist 0.4314, final 0.4331.
- PM3 under the same recipe is much worse: best 0.7039 before early stopping; Chamfer fails badly from epoch 0.

PM3 Best Recipe Without Early Stopping



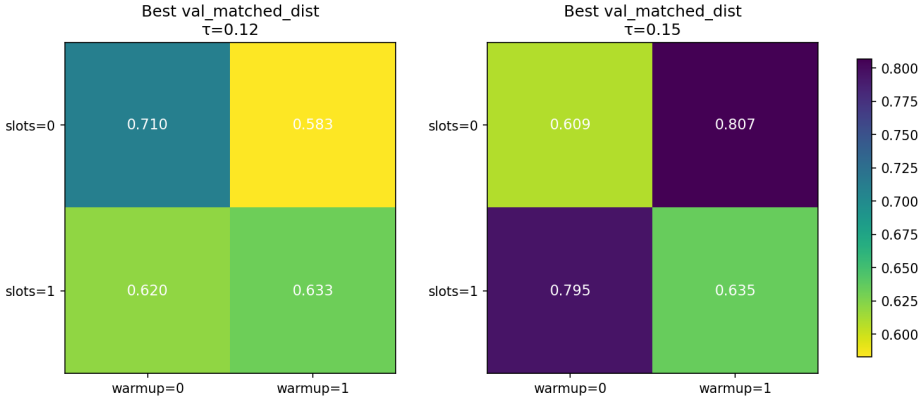
- Early stopping was a real confounder: best dist improves from 0.7039 to 0.6607 without truncation.
- But even the full 200-epoch rerun still lags the best single-factor PM3 variants.
- Conclusion: early stopping hurt this run, but it is not the whole explanation.

PM3 Recipe Comparison



- The best PM3 points are warmup-only (0.5495), tau-only (0.5625), and the grid winner (0.5828).
- The full $\tau = 0.15$ + slots + warmup recipe is worse in both the grid rerun (0.6353) and the no-early-stop rerun (0.6607).
- This is strong evidence that the PM3 gains are non-additive.

2x2x2 PM3 Grid

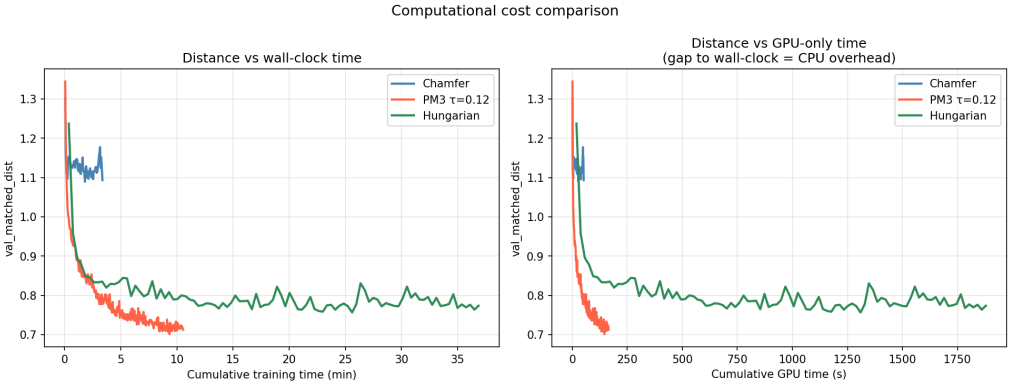


- Best clean PM3 recipe in the grid: $\tau = 0.12$ + warmup + standard decoder at 0.5828.
- Slots help at $\tau = 0.12$ without warmup (0.6199), but stop helping once warmup is added.
- At $\tau = 0.15$, warmup actually hurts the standard decoder badly ($0.6090 \rightarrow 0.8069$).

Why The Combined Recipe Is Not Additive

- Slots likely help PM3 by breaking symmetry when the decoder is otherwise too exchangeable.
- warmup_cosine also smooths optimization, and a hotter τ softens responsibility assignment.
- Combining all three can over-smooth the PM3 objective: coverage is maintained, but the push toward sharp, consistent assignments weakens.
- Hungarian benefits from the same architectural and schedule improvements because its assignment step stays discrete.
- The gap between the grid rerun and the separate no-early-stop rerun also suggests narrower, more brittle optimization basins for the combined PM3 recipe.

Computational Cost Comparison



- PM3 reaches 0.80 in 2.82 min; Hungarian reaches it in 5.93 min; Chamfer never does.
- PM3 and Chamfer both run around 6 ms/batch; Hungarian is about 139 ms/batch.
- Hungarian can still reach a better final distance than default PM3 only when given the stronger combined recipe.

Benchmark Validity

- This CLEVR study was very effective for parameter search and mechanism analysis.
- It is not the final community benchmark story: the current task is a state-denoising proxy with matched-distance evaluation, not the most popular image-to-set benchmark in the loss/matching literature.
- For a mainstream '$X \to \mathrm{set}$' benchmark with a rich assignment literature, COCO detection in a DETR-style setup is the right next target.
- So the correct framing is:
  - CLEVR = fast studybed to find promising PM3 settings
  - COCO DETR = benchmark to test whether PM3-like matching holds up in a widely studied setting

Confirmed Next Steps

- Keep the CLEVR study outputs, slides, scripts, and utilities in a dedicated study area instead of scattered repo roots.
- Carry forward two PM3 candidates: the strongest clean grid point and the best single-factor PM3 wins.
- Bootstrap a dedicated COCO DETR benchmark area with:
  - a canonical Hungarian DETR baseline
  - a PM3-like soft matching variant on the same class + box cost matrix
  - reduced-data smoke tests before full COCO training
- Compare AP, convergence speed, duplicate behavior, and training stability before making broader claims.